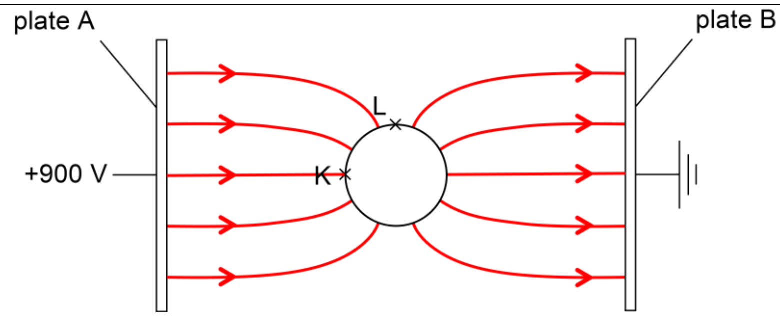
Qn	**Suggested Solutions**
2(a)	Coulomb's law states that the electric force acting between any two point charges is • directly proportional to the product of the charges and • inversely proportional to the square of their distance apart
(b)	Electric field strength (at a point in an electric field) is equal to the negative electric potential gradient (at that point)
(c)	For potential to be zero, one potential must be positive and the other potential must be negative OR For potential to be zero, the charges must have opposite sign
	For field to be zero, the fields (due to X and Y) must be in opposite directions OR For field to be zero, the charges must have the same sign
	The signs of the charges cannot (simultaneously) be both the same and opposite (so not possible)
(d)(i)	
	• At least six field lines drawn with the • field lines equally spaced near the plates. • Field lines are outside the sphere. • Arrows indicating the direction of the electric field is from the higher potential surface to the lower potential surface (towards the right) • 90° to the surfaces on the sphere and the plates.
(ii)	Charges are free to move in a conductor. In electrostatic equilibrium/for no net movement of charges,
	The component of electric field parallel to the surface of the sphere must be zero. Hence, the potential difference at points on the surface of the charge is zero and the potentials at K and L are equal.
(iii)	$V_K = \frac{900}{2}$ $= 450$ V